

LANZHOU, China

WMO: 528890

Lat: 36.050N

Lon: 103.883E

Elev: 1518

StdP: 84.37

Time Zone: 8.00 (E08)

Period: 90-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-9.8	-19.9	0.8	-4.1	-18.4	0.9	-4.5	3.4	-1.9	3.0	-1.7	0.5	90	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	32.6	18.0	31.1	17.4	29.7	16.8	20.3	28.6	19.5	27.5	18.7	26.5	2.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	15.4	24.6	16.9	14.5	23.5	16.1	13.7	22.6	66.2	28.5	63.1	27.4	60.1	26.7	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.3	3.5	3.1	DB	-13.8	35.7	2.1	1.9	-15.3	37.1	-16.5	38.2	-17.7	39.2	-19.2	40.6	(4)
(5)			WB	-14.7	21.7	1.9	0.8	-16.0	22.3	-17.1	22.7	-18.2	23.1	-19.5	23.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-3.5	1.2	6.9	13.1	17.7	21.5	23.5	22.2	17.9	11.1	4.2	-2.1	(6)	
	DBStd	9.80	2.55	3.17	3.83	4.21	3.46	2.65	3.00	2.84	3.00	3.12	3.29	2.60	(7)	
	HDD10.0	1371	418	247	109	17	2	0	0	0	0	26	176	376	(8)	
	HDD18.3	3064	677	481	353	164	52	6	1	3	42	226	425	634	(9)	
	CDD10.0	1799	0	0	14	109	239	344	418	377	238	59	1	0	(10)	
	CDD18.3	451	0	0	0	6	32	100	161	122	30	0	0	0	(11)	
	CDH23.3	3822	0	0	2	83	307	833	1439	952	204	2	0	0	(12)	
	CDH26.7	1278	0	0	0	20	63	252	584	313	47	0	0	0	(13)	
(14) Wind	WSAvg	0.9	0.4	0.7	1.1	1.2	1.3	1.3	1.2	1.0	0.8	0.6	0.4	0.4	(14)	
(15) Precipitation	PrecAvg	317	1	2	8	18	38	40	64	72	47	24	4	1	(15)	
	PrecMax	536	8	9	23	68	126	109	142	155	92	60	21	6	(16)	
	PrecMin	189	0	0	0	1	1	12	14	7	9	0	0	0	(17)	
	PrecStd	75	2	2	6	14	30	23	32	38	24	18	5	1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	13.7	22.0	29.6	30.5	33.1	36.0	33.8	31.0	22.8	16.1	7.0	(19)	
		MCWB	0.4	4.4	9.5	13.1	13.8	15.7	19.6	18.9	15.9	12.4	7.7	1.0	(20)	
	2%	DB	4.9	11.2	18.6	25.6	28.5	31.0	33.2	31.6	27.7	20.7	14.1	5.3	(21)	
		MCWB	-1.2	3.4	7.7	11.3	13.5	16.3	18.7	18.5	16.2	11.5	6.5	-0.1	(22)	
	5%	DB	3.0	8.9	16.3	23.0	26.7	29.5	31.6	29.9	25.5	18.7	12.1	4.0	(23)	
		MCWB	-2.1	2.1	6.6	10.2	12.9	15.6	18.3	18.0	15.5	10.7	5.4	-1.0	(24)	
	10%	DB	1.4	7.0	14.0	20.7	24.9	28.0	30.1	28.3	23.7	16.9	10.1	2.7	(25)	
		MCWB	-3.1	0.9	5.5	9.4	12.4	15.2	17.8	17.6	15.0	9.8	4.5	-1.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	5.2	10.1	14.4	16.2	19.1	21.7	21.3	20.4	14.1	8.6	1.6	(27)	
		MCDB	6.8	12.3	20.1	23.7	24.6	27.2	32.5	29.8	24.3	20.0	15.1	6.4	(28)	
	2%	WB	-1.0	3.6	8.4	12.4	15.1	17.9	20.5	20.2	17.7	12.5	7.0	0.1	(29)	
		MCDB	4.2	10.3	16.4	22.4	24.3	26.0	30.0	27.7	24.4	18.8	12.8	4.9	(30)	
	5%	WB	-2.0	2.4	7.2	11.2	14.2	17.2	19.7	19.5	16.6	11.6	5.9	-0.8	(31)	
		MCDB	2.8	8.3	14.7	21.1	23.3	25.6	28.4	26.9	23.2	17.1	11.5	3.8	(32)	
	10%	WB	-3.0	1.2	6.0	10.1	13.4	16.5	18.9	18.7	15.6	10.6	4.7	-1.8	(33)	
		MCDB	1.1	6.6	13.5	19.2	22.6	24.8	27.4	26.0	22.3	15.6	9.6	2.4	(34)	
(35) Mean Daily Temperature Range	MDBR		10.2	10.7	10.8	11.5	11.2	11.0	10.5	10.4	10.1	10.2	10.4	9.7	(35)	
	5% DB	MCDBR	12.6	13.9	15.2	15.7	14.8	14.1	13.3	13.0	13.6	13.1	13.4	11.0	(36)	
		MCWBR	8.4	8.1	7.4	6.1	5.2	4.2	3.6	4.1	4.9	5.9	7.5	7.3	(37)	
	5% WB	MCDBR	12.3	13.0	13.6	13.3	12.2	11.2	11.1	10.7	10.7	11.8	12.6	10.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.369	0.432	0.499	0.538	0.534	0.520	0.501	0.514	0.489	0.452	0.384	0.352	(40)	
	taud		2.044	1.893	1.765	1.719	1.800	1.894	2.017	1.956	1.967	1.977	2.070	2.119	(41)	
	Ebn at Noon		790	778	761	758	769	779	791	770	764	749	767	785	(42)	
	Edh at Noon		128	168	209	231	216	197	173	180	169	153	124	111	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.93	3.71	4.72	5.57	5.95	5.88	5.73	5.27	4.05	3.69	3.13	2.80	(44)	
	RadStd		0.28	0.26	0.31	0.33	0.35	0.34	0.53	0.37	0.30	0.34	0.24	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+32	N/A

Nomenclature: See separate page